

Quick check exercises 5.2

1. Definition of an antiderivative

A function F is an antiderivative of a function f on an interval if

$$F'(x) = f(x)$$

for all x in the interval.

Answer:

$$F'(x) = f(x)$$

2. Write an equivalent integration formula

Remember:

If

$$\frac{d}{dx}[F(x)] = f(x),$$

then

$$\int f(x) dx = F(x) + C.$$

where C is the constant of integration.

(a)

Given:

$$\frac{d}{dx}[\sqrt{x}] = \frac{1}{2\sqrt{x}}$$

Therefore,

$$\int \frac{1}{2\sqrt{x}} dx = \sqrt{x} + C$$

(b)

Given:

$$\frac{d}{dx}[e^{4x}] = 4e^{4x}$$

Therefore,

$$\int 4e^{4x} dx = e^{4x} + C$$

3. Evaluate the integrals

(a)

Evaluate:

$$\int (x^3 + x + 5) dx$$

Integrate term by term:

$$\int x^3 dx = \frac{x^4}{4},$$

$$\int x dx = \frac{x^2}{2},$$

$$\int 5 dx = 5x.$$

Adding them together:

$$\frac{x^4}{4} + \frac{x^2}{2} + 5x + C$$

(b)

Evaluate:

$$\int (\sec^2 x - \csc x \cot x) dx$$

Recall:

$$\frac{d}{dx}(\tan x) = \sec^2 x,$$

and

$$\frac{d}{dx}(\csc x) = -\csc x \cot x.$$

Thus,

$$\int \sec^2 x dx = \tan x,$$

and

$$\int (-\csc x \cot x) dx = \csc x.$$

Therefore,

$$\tan x + \csc x + C$$

4. Integral curve problem

The graph

$$y = x^2 + x$$

is an integral curve for the function f .

An integral curve means

$$f(x) = y'.$$

Differentiate:

$$y' = \frac{d}{dx}(x^2 + x) = 2x + 1.$$

Therefore,

$$f(x) = 2x + 1$$

Now, if another integral curve G satisfies

$$G'(x) = 2x + 1,$$

then

$$G(x) = x^2 + x + C.$$

Given

$$G(1) = 5,$$

we have

$$1^2 + 1 + C = 5,$$

$$2 + C = 5,$$

so

$$C = 3.$$

Hence,

$$G(x) = x^2 + x + 3.$$

Finally,

$$G(2) = 2^2 + 2 + 3 = 4 + 2 + 3 = 9.$$

Answers:

$$f(x) = 2x + 1$$

$$G(2) = 9$$

5. Slope field

The differential equation is

$$\frac{dy}{dx} = \frac{2x}{x^2 - 4}.$$

The slope at a point depends only on its x -coordinate.

Through $(0, 5)$

Substitute $x = 0$:

$$\frac{dy}{dx} = \frac{2(0)}{0^2 - 4} = 0.$$

Slope:

$$0$$

Through $(-4, 1)$

Substitute $x = -4$:

$$\frac{dy}{dx} = \frac{2(-4)}{(-4)^2 - 4} = \frac{-8}{16 - 4} = \frac{-8}{12} = -\frac{2}{3}.$$

Slope:

$$-\frac{2}{3}$$

Final Answers

1.

$$F'(x) = f(x)$$

2. (a)

$$\int \frac{1}{2\sqrt{x}} dx = \sqrt{x} + C$$

(b)

$$\int 4e^{4x} dx = e^{4x} + C$$

3. (a)

$$\frac{x^4}{4} + \frac{x^2}{2} + 5x + C$$

(b)

$$\tan x + \csc x + C$$

4.

$$f(x) = 2x + 1, \quad G(2) = 9$$

5. Slopes: $0, -\frac{2}{3}$

EXERCISE SET 5.2

1. Confirm the derivative formula and state the corresponding integration formula

(a)

Given:

$$\frac{d}{dx} \left[x \sqrt{1+x^2} \right] = \frac{x}{\sqrt{1+x^2}}$$

Check by differentiating

Let

$$F(x) = \sqrt{1+x^2} = (1+x^2)^{1/2}.$$

Using the chain rule,

$$F'(x) = \frac{1}{2}(1+x^2)^{-1/2}(2x) = \frac{x}{\sqrt{1+x^2}}.$$

The stated derivative is correct. 

Corresponding integration formula

Since

$$\frac{d}{dx} \left[\sqrt{1+x^2} \right] = \frac{x}{\sqrt{1+x^2}},$$

we have

$$\int \frac{x}{\sqrt{1+x^2}} dx = \sqrt{1+x^2} + C.$$

(b)

Given:

$$\frac{d}{dx} [xe^x] = (x+1)e^x$$

Check by differentiating

Use the product rule:

If

$$u = x, \quad v = e^x,$$

then

$$\frac{d}{dx}(uv) = u'v + uv'.$$

Thus,

$$\frac{d}{dx}(xe^x) = (1)(e^x) + x(e^x) = e^x + xe^x = (x+1)e^x.$$

The formula is correct. 

Corresponding integration formula

Therefore,

$$\int (x+1)e^x dx = xe^x + C.$$

2. Confirm that the stated integration formula is correct by differentiating

(a)

Given:

$$\int x \sin x dx = \sin x - x \cos x + C.$$

Differentiate the right-hand side:

$$\frac{d}{dx}(\sin x - x \cos x).$$

Using the product rule on $x \cos x$,

$$\frac{d}{dx}(x \cos x) = \cos x - x \sin x.$$

Therefore,

$$\frac{d}{dx}(\sin x - x \cos x) = \cos x - (\cos x - x \sin x).$$

Simplifying,

$$= \cos x - \cos x + x \sin x = x \sin x.$$

This matches the integrand.

Hence,

$$\int x \sin x \, dx = \sin x - x \cos x + C$$

is correct. ✓

(b)

Given:

$$\int \frac{dx}{(1-x^2)^{3/2}} = \frac{x}{\sqrt{1-x^2}} + C.$$

Differentiate

$$F(x) = \frac{x}{\sqrt{1-x^2}} = x(1-x^2)^{-1/2}.$$

Using the product rule,

$$F'(x) = (1)(1-x^2)^{-1/2} + x \left(-\frac{1}{2}\right) (1-x^2)^{-3/2}(-2x).$$

Simplify:

$$= (1-x^2)^{-1/2} + x^2(1-x^2)^{-3/2}.$$

Factor out

$$\begin{aligned} & (1-x^2)^{-3/2} : \\ &= (1-x^2)^{-3/2} [(1-x^2) + x^2]. \end{aligned}$$

Since

$$(1-x^2) + x^2 = 1,$$

we get

$$F'(x) = (1-x^2)^{-3/2} = \frac{1}{(1-x^2)^{3/2}}.$$

Thus the formula is correct. ✓

3. Focus on Concepts

What is a constant of integration?

A **constant of integration** is the arbitrary constant C added to an indefinite integral.

For example,

$$\int 2x \, dx = x^2 + C.$$

The constant appears because differentiation eliminates constants:

$$\frac{d}{dx}(x^2 + 5) = 2x, \quad \frac{d}{dx}(x^2 - 3) = 2x.$$

Both functions have the same derivative.

Why does an answer to an integration problem involve a constant?

Integration finds **all antiderivatives** of a function.

Since infinitely many functions differ only by a constant and have the same derivative, we include

$$\boxed{+C}$$

to represent the entire family of antiderivatives.

4. Focus on Concepts

What is an integral curve of a function f ?

An **integral curve** of a function f is a graph of a function $y = F(x)$ such that

$$F'(x) = f(x).$$

In other words, its slope at every point equals the value of $f(x)$.

How are two integral curves of the same function related?

If F and G are both integral curves of f , then

$$F'(x) = G'(x) = f(x).$$

Therefore,

$$F(x) - G(x) = C,$$

where C is a constant.

So, **two integral curves of the same function are vertical translations of each other**—they have the same shape and differ only by a constant amount.

Final Answers

1.

(a)

$$\int \frac{x}{\sqrt{1+x^2}} dx = \sqrt{1+x^2} + C$$

(b)

$$\int (x+1)e^x dx = xe^x + C$$

2.

Both formulas are correct because differentiating their right-hand sides gives the original integrands.

3.

A constant of integration is the arbitrary constant C added to represent all antiderivatives, since derivatives of constants are zero.

4.

An integral curve is a function whose derivative equals f . Any two integral curves of the same f differ by a constant and are vertical shifts of one another.